

Managing Task Interdependencies in Multi-Team Projects: A Longitudinal Study

Martin Hoegl and Katharina Weinkauff

*WHU – Otto Beisheim Graduate School of Management, Vallendar, Germany;
Technical University of Berlin*

ABSTRACT In this article we examine project-level and team-level managerial functions aimed at managing inter-team task interdependencies and investigate their effect on the performance of teams in a multi-team product development project. We hypothesize that team interface management (a team-level function) performed in the concept phase of a project, rather than the later development phase, is positively related to team performance at the project's conclusion (i.e. product quality, product cost, development budget, development time). Furthermore, we argue that project structuring and support (a project-level function) is important in both the concept and the development phases. We test our hypotheses empirically based on a 36 months longitudinal study of a product development project in the European automotive industry comprising 39 teams. The results show that team interface management is particularly important during the concept phase of the project. Project structuring and support is most important in the development phase of the project, while hindering team performance in the concept phase. Theoretical and practical implications are discussed.

INTRODUCTION

Facing increasing globalization and competition combined with rapidly changing technologies, companies must develop innovative products in a short period of time to become and remain competitive. For new product development efforts, project teams have been widely used and proved to be generally effective (Ancona and Caldwell, 1992; Brown and Eisenhardt, 1995; Clark and Fujimoto, 1991; Cooper and Kleinschmidt, 1994; Denison et al., 1996; Dougherty, 1992; Hoegl and Gemuenden, 2001; Mohrman et al., 1995). Furthermore, researchers and

Address for reprints: Martin Hoegl, WHU – Otto Beisheim Graduate School of Management, Chair of Leadership and Human Resource Management, Burgplatz 2, 56179 Vallendar, Germany (hoegl@whu.edu).

practitioners agree that effective management plays a critical role in ensuring project success (Cleland and King, 1988; Pinto and Prescott, 1988).

However, these concepts and findings largely refer to smaller-scale projects consisting of a single project team. Responding to advances in science and technology as well as the increasing demands from customers for integrated and 'full-featured' solutions, products are becoming more and more complex. In order to achieve high quality innovations in a satisfactory time-to-market, hundreds of people may be required to develop components of a new product simultaneously. In these cases, companies tend to create multi-team projects, i.e. large-scale projects consisting of several teams with separate tasks working interdependently on the same overall project (Cusumano and Selby, 1995; Sabbagh, 1996). With the project organization typically corresponding to the product architecture, teams are charged with the development of a single module of the product, and interdependencies between the teams reflect the technical interfaces of the modules (Brusoni and Prencipe, 2001; Gerwin and Moffat, 1997; Kazanjian et al., 2000; Sanchez and Mahoney, 1996; von Hippel, 1990). Moreover, contrary to smaller-scale projects generally consisting of a team and a project manager supervising the team, management responsibilities are significantly more complex in multi-team projects. The project architecture often features a hierarchy of teams, with cross-functional teams as the primary performance units on the lower level and a team of project managers on the upper level. Teams and project managers involved in these innovative settings are challenged by a highly complex organizational context with intricate tasks, strong interdependence, and time urgency (Gerwin and Moffat, 1997; Zirger and Hartley, 1994).

Researchers have begun to investigate large-scale projects more specifically and thereby focused on selected aspects such as authorizing processes (Gerwin and Moffat, 1997), learning and creativity (Kazanjian et al., 2000), knowledge specialization (Brusoni et al., 2001), as well as task partitioning (von Hippel, 1990) and coordination (Sanchez and Mahoney, 1996). However, while such previous research has identified various sources of task interdependencies in large-scale projects, such prior work has only begun addressing (both theoretically and empirically) the issue of managing interdependencies in multi-team product development projects. Managing the intense task interdependencies, however, is likely critical to the success of multi-team projects, as the work and output of any single team has consequences for the work and outputs of other teams in the project. For instance, a team's technical decisions (e.g. regarding weight, dimensions, or other properties of a team's module) or any schedule slippage immediately affect other teams and the entire project (Gerwin and Moffat, 1997).

Thus, while the necessity to carefully manage inter-team task interdependencies in such projects seems important, questions remain as to who in the project organization should manage these task interdependencies, and in what phase of the project it is most critical to do so. In this research we address these two questions by

distinguishing the management of task interdependencies hierarchically in the multi-team organization (i.e. the team level and the superordinate project level) as well as temporally (Mitchell and James, 2001) over the course of the project (i.e. the concept phase and the later development phase). We provide empirical evidence for our hypotheses using data from a 36 months longitudinal study of a product development project in the European automotive industry comprising 39 teams.

This study attends to significant gaps in the literature. First, as discussed above, there is a lack of conceptual and empirical work providing recommendations on managing the strong task interdependencies in multi-team product development projects. This research specifically addresses the roles that project managers and the teams themselves play in managing task interdependencies in multi-team projects, extending previous research on single-team projects as well as new product development (Ancona and Caldwell, 1990, 1992; Gersick, 1988; Katz, 1982). Second, prior contributions to the literature on managerial functions in project teams have either failed to recognize time as a critical dimension in the successful application of managerial activity (Mitchell and James, 2001) or have roundly asserted the early concept phase as particularly critical without providing the necessary empirical evidence (Gales et al., 1992; Kazanjian et al., 2000; Thamhain and Wilemon, 1975). In this study, we focus on the concept phase and the development phase of a large-scale product development project and investigate how project- and team-level managerial functions during these two phases affect the teams' performance at the project's conclusion. Lastly, large-scale product development projects, which typically occur in such industries as automotive or aerospace, as well as in the field of IT software and hardware, are not only prestigious for companies and their executives, but also of considerable importance to an organization's financial wellbeing and strategic competitiveness. Furthermore, myriad private and business consumers usually rely on these large, major products (Cusumano and Selby, 1995). Therefore, learning about success factors of large-scale product development projects is important to practice and research (Morris and Hough, 1987).

We begin the detailed discussion of our conceptual arguments by specifying relevant dimensions of team performance as well as team-level and project-level managerial functions related to the management of task interdependencies in multi-team product development projects. Lastly, we identify how these managerial functions matter in different project phases and summarize our predictions in the form of hypotheses.

CONSTRUCTS AND HYPOTHESES

Team Performance

The conceptualization of team performance as a multi-dimensional construct is widely acknowledged in the literature (Denison et al., 1996; Hoegl and

Gemuenden, 2001). In general, team performance can be defined as the extent to which a team is able to meet established objectives. In product development projects, specific dimensions of team performance include the adherence to predefined product quality and cost (i.e. effectiveness) as well as development budget and time (i.e. efficiency) objectives.

In the context of this study, product quality refers to certain desired properties of the module developed by the team. For a product development team charged with designing a specific module of a larger product, several properties may be important including functionality, manufacturability, durability and robustness, dimensional integrity, as well as optical and tactile attractiveness. The teams are charged, however, with achieving these qualitative properties within the constraints of targeted module costs (i.e. total anticipated cost for the module) and can be assessed on their ability to meet these cost targets. Adherence to development budget objectives refers to the costs associated with the team's development activities (i.e. personnel, prototype material, testing, etc). As for schedule objectives, all teams in a multi-team project are included in an overall sequence of milestones where design reviews take place and certain deliverables are expected at predefined times, which in turn provide necessary input for other teams.

Team-Level and Project-Level Management of Task Interdependencies

The focus of this study is on management in multi-team product development projects, and here specifically on the performance of those acts which are critical to managing the interplay between interdependent teams belonging to the same project. The underlying argument of this paper is that the management of multi-team interdependencies in large-scale projects is not concentrated on project-level managers or the teams, but that both perform necessary managerial functions.

As emphasized by many researchers (Cusumano and Selby, 1995; Gerwin and Moffat, 1997; Kazanjian et al., 2000; Mohrman et al., 1995; Morris and Hough, 1987), in large-scale projects there is a need for a project management level responsible for the total product, which usually consists of a team of project managers. This literature concludes that the fundamental purpose of project managers leading a multi-team project is to achieve integration among the module teams and thereby accomplish product integrity. An important means for doing so is to establish a supportive organizational context in which the teams are embedded (Cusumano and Selby, 1995; Mohrman et al., 1995; Morris and Hough, 1987). We refer to this project-level managerial function as *project structuring and support*, entailing several elements that are aimed at managing task interdependencies (Kazanjian et al., 2000). The project managers establish and maintain a project architecture often modelled after the product architecture, and clarify every team's position and boundaries within this structure (Brusoni et al., 2001; Gerwin and

Moffat, 1997; Sanchez and Mahoney, 1996; von Hippel, 1990). As such, identifying interfaces and clarifying boundaries and responsibilities between the teams in the project helps reduce uncertainty and complexity inherent in the development process (Mohrman et al., 1995). Furthermore, the project managers define and communicate the responsibilities and decision authorities for each team and provide the necessary resources (e.g. budget, human resources, etc). Adequate resources are important for the teams to have the means to respond to task requests of related teams, such as change requests following necessary changes on their (interdependent) modules.

While the project managers usually provide the teams with the structure of the project (including an initial description of the task interdependencies between the teams), the teams themselves are asked to put this structure into action. Interdependencies between teams require an exchange of information, products, and services. Especially, design information need to be transmitted between teams to make sure the different modules fit within the broader system (Kazanjian et al., 2000). In order to better achieve transparency of the varied interrelationships, teams must internalize their relevant interfaces (Gerwin and Moffat, 1997; Mohrman et al., 1995). We term this team-level interdependence management function *team interface management*, entailing the technical definition of the team's module and its technical input and output parameters, as well as the coordination of those details with technically related (i.e. interdependent) teams. The work of Ancona and Caldwell (1990, 1992) and Mohrman et al. (1995) underlines that teams need to manage their relationships with interdependent teams. These authors show that the management of interfaces is critical for the success of single-team projects. In this research we argue that this is particularly true for teams in innovative multi-team projects. Given the intense task interdependencies between teams designing and developing related modules, the success of any one team is dependent upon how well their work results integrate with that of related teams. For instance, a team responsible for designing and developing the exhaust system of an automobile must precisely clarify its technical interfaces (e.g. technical input and output parameters) with the engine teams, the chassis teams, and others to ensure that the exhaust system will, in the end, fit in its space, can be connected properly to the engine and chassis, and so on.

As such, both project structuring and support as well as team interface management are important for teams to be effective and efficient. A proper understanding of the network of teams (i.e. defining communication lines, responsibilities, and expectations between the teams) facilitates inter-team coordination and communication and hence operational efficiency in terms of development time and cost (Ancona and Caldwell, 1992; Barczak and Wilemon, 1989; Steckler and Fondas, 1995). Initiating and maintaining contact with other teams and external groups and securing effective transactions and information flows is crucial for teams to design and develop modules that adhere to predefined qualitative expect-

tations (e.g. functionality, durability, dimensions, etc) while staying within targeted module cost limits (Ancona and Caldwell, 1988, 1992; Barczak and Wilemon, 1991; Cartwright and Zander, 1968; Gladstein, 1984; Jassawalla and Sashittal, 2000; Mohrman et al., 1995). Moreover, it is important for the self-identification and cohesion of the teams to delineate their boundaries and determine their role in the project (Ancona, 1990; Ancona and Caldwell, 1988; Alderfer, 1987; Mohrman et al., 1995). Clear boundaries provide some protection to a team and help it establish an identity and ability to integrate and work as a unit (Gerwin and Moffat, 1997; Scott, 1997).

Managing Task Interdependencies in Different Project Phases

As the technical nature of the new product as well as the project context changes over time, so may the necessity to manage task interdependencies within the project (Pinto and Prescott, 1988). Given the technical uncertainty and complexity of large-scale product development projects (e.g. a once favoured technical concept for a module proves unfeasible during testing) as well as market changes (e.g. a competitor announces a rival product offering new features not yet included in this product's design), both project-level and team-level action is required to adapt to the new situation in terms of project structure, budget allocations, redefinition of technical interfaces, etc. As such, neither project structuring and support nor team interface management are functions that are solely performed at the beginning of the project, but throughout the life of the project to adapt to new requirements as the project unfolds.

Phases of product development projects. Researchers agree that 'as a project proceeds through its life cycle, it passes through an identifiable sequence of phases, distinguished from each other by the type of task characteristic of each phase . . .' (Adams and Barndt, 1988, p. 209). There are between three and six project phases distinguished in the literature (Adams and Barndt, 1988; Pinto and Prescott, 1988). Although there is no common terminology, the basic idea is that the beginning phases are characterized by planning, conceptualization, creation, formation, build-up, idea generation, and design. The middle phases are termed development, execution, main program, or problem solving. A project ends with a phase of commercialization, termination, diffusion, ending, phase-out, implementation, or stabilization (Adams and Barndt, 1988; Ancona and Caldwell, 1990; Cusumano and Selby, 1995; Gales et al., 1992; Pinto and Prescott, 1988; Thamhain and Wilemon, 1975). Following Ancona and Caldwell (1990), we divide the product development process into three identifiable phases, which we name 'concept', 'development', and 'pre-production' phase. In the concept phase, the project is initiated, goals are set, priorities are established, the project is designed, and the task is structured. The aim is to move from a more general idea to a specific design

plan. In the development phase, the actual work is done on the module designs the teams have chosen, and technical problems are solved. The project teams are 'under the gun' to meet their goals regarding schedule, budget, and quality specifications (Adams and Barndt, 1988). The pre-production phase is characterized by transferring ownership of the product to manufacturing and sales, and finally to the product's intended users (Pinto and Prescott, 1988). As Gales et al. (1992) point out, these project phases are not entirely distinct, and activities characterizing one phase do not necessarily cease in another. It is merely the primary functions and the focus of a project that shift sequentially.

Project structuring and support in different project phases. The project organization is mainly shaped in the first phase, i.e. concept phase, of a project (Gales et al., 1992; Kazanjian et al., 2000). Accordingly, researchers have argued that it is in this early phase that the responsibilities and resources of the module teams are determined (Ancona and Caldwell, 1990), and that potential administrative disagreements are 'preempted', in order to avoid an escalation of these issues in the more advanced project phases (Thamhain and Wilemon, 1975). However, the design of the product – and the project – cannot be perfectly determined at the outset of an innovative process. Due to task interdependencies, changes that occur in one team might affect the whole project and necessitate different project configurations and resource allocations. Changes, however, can become necessary particularly during the development phase of the project, as module concepts are turned into early prototypes, which are subsequently tested individually and collectively to ensure proper technical integration of all modules. In that phase, teams fine-tune their modules interactively with related modules and hence need to negotiate technical issues as they emerge. During this process, project managers need to foster communication between teams as well as again clarify, and where necessary adapt, the organizational structure and resource allocations to changing technical realities as the project unfolds (Kazanjian et al., 2000). Therefore, we argue that project structuring and support is important in both the concept and the development phases. Hence, we summarize as follows:

Hypothesis 1a: Project structuring and support as performed in the concept phase and in the development phase is positively associated with team performance (product quality, product cost, adherence to development budget, adherence to development schedule) in multi-team product development projects.

Hypothesis 1b: Project structuring and support is performed equally in the concept and in the development phases of the project.

Team interface management in different project phases. Developing an appropriate boundary for the team and structuring its relationship with other teams working on

related modules is essential in the first project phase (Ancona and Caldwell, 1990; Gersick, 1988; Katz, 1982). This helps create a more workable level of certainty and clarity as the teams identify technically related teams and individual counterparts within those teams in order to effectively and efficiently communicate ideas and concepts. As such, team interface management in the concept phase of the project is particularly necessary as the teams themselves are dependent upon information from related teams to ensure that their module concepts technically integrate with related modules. In the development phase, maintaining relationships with other teams is still important, however, the teams' focus must shift to internal problem solving and proceeding with development work in order to meet project deadlines and milestones. As such, the end of the project's concept phase marks a point where the teams finalize their module design planning and come to a 'concept freeze' providing a basis for subsequent development work. Unless there are radical changes, inputs from interface teams should be limited. Otherwise, the team might fail to move on. Hence, we do not argue that engaging in team interface management activities during the development phase of a project will be related to better performance at the project's conclusion. In summary, we hypothesize:

Hypothesis 2a: Team interface management as performed in the concept phase is positively associated with team performance (product quality, product cost, adherence to development budget, adherence to development schedule) in multi-team product development projects.

Hypothesis 2b: Team interface management is performed to a greater degree in the concept than in the development phases of the project.

METHODS

Research Setting

This study was conducted in a new product development project in the European automotive industry. The new product was an innovative addition to the models the firm already produced, and was intended to ensure competitiveness and growth. The total budget for this development project was approximately US\$500 million. The project we investigated included a total of 39 cross-functional teams that were co-located in one building. They consisted of members from different functions, such as engineering, quality management, controlling, purchasing, logistics, marketing, planning, etc, and had an average size of nine team members. Some teams also included employees from supplier organizations; however, they acted as consulting team members rather than core team members as they did not participate in daily work. Therefore, we excluded them from data gathering.

The 39 teams all had formal (team-internal) team leaders and were divided up into eight sub-projects (such as powertrain, interior, electronics, etc) each headed by a project manager, who in turn reported to the project director responsible for the entire project. The project managers' job was to make sure that the overall product specifications were met. Therefore, they continuously monitored the progress of the teams and the overall project. To this end, project managers set specific project milestones, at which design reviews were conducted.

Data Collection

Data were gathered by the authors and research assistants in individual interviews using a fully standardized questionnaire (five point answer scale). All interviews were conducted on site in dedicated interview rooms assuring similar conditions for each interview. The interviews followed a very structured pattern. After a brief introduction of the study, the questionnaire was handed over to the respondent to complete by reading it himself or herself. This way, possible interviewer effects were minimized while there was still an interviewer present to clarify questions if any occurred. Following the completion of the questionnaire, the respondents were invited to share any additional open-ended comments, which the interviewer noted on paper. Respondents' participation in this study was strictly voluntary. All contacted respondents were interviewed.

The data were gathered at three times in the 36 months duration of the project. The first data collection (time 1) was conducted in a time span of about six weeks in months 12 and 13 of the project, followed by a second round of data collection (time 2) in months 23 and 24. A third round of data collection (time 3) took place in month 36 gathering team performance ratings from project managers, quality experts, and company records. These data collection times were chosen in accordance with the project's main milestones, where time 1 marked the conclusion of the concept phase, time 2 the conclusion of the development phase, and time 3 the conclusion of the pre-production phase.

The database for this research is made up of 452 interview responses (time 1 and time 2) from team members and team leaders, expert ratings of product quality as well as manager ratings of schedule performance, and data from company records regarding product cost and development cost (time 3). It should be noted that, at time 3, company records regarding product cost at the module level (i.e. team level) were available for 27 of 39 teams. Controlling product costs at the overall project level, the company did not maintain specific product cost controlling data for the other twelve teams, as their modules were regarded as less cost-sensitive. Furthermore, missing data (due to partial non-response on questionnaires) reduces the team-level sample for the variables project structuring and support ($N = 37$ at time 1 and time 2), team interface management ($N = 38$ at time 2), product quality ($N = 38$), and weeks beyond schedule ($N = 37$).

To ensure content validity and avoid common source bias, we used data from different raters and sources to measure the different constructs investigated in this research. Team interface management was measured using data from multiple team members, project structuring and support was assessed with team leader responses, while the team performance dimensions were measured using company records, quality expert ratings, as well as project manager evaluations of the teams' adherence to schedule.

Measures

Due to the lack of previously existing measurement scales in the context of multi-team projects, we developed the scales for use in this study based on the discussion and measures of related constructs in the literature (Ancona and Caldwell, 1992; Cusumano and Selby, 1995; Mohrman et al., 1995). The questionnaire was administered in the language of the country where the research setting was situated. Translations of the items are provided in the appendix of this article. Identical measures were used for assessing management functions at time 1 and time 2. At time 2, the respondents were instructed to refer to the time period since the first data collection in answering the questions regarding project structuring and support as well as team interface management.

Largely building on the descriptions of project managers' tasks provided by Cusumano and Selby (1995) and Mohrman et al. (1995), we developed five items for assessing project structuring and support. Team leaders rated the extent to which they perceived the project management team exercised this function in the previous project phase. Reliability analysis showed a satisfactory level of internal consistency with a Cronbach's alpha coefficient of 0.82.

For measuring team interface management, we mainly relied on Mohrman et al.'s (1995) discussion of team management tasks in team-based organizations and on Ancona and Caldwell's (1992) description of boundary management. Team members were asked to evaluate the extent to which the activities described in the seven items were performed in the team. Prior to aggregating team members' evaluations of team interface management, interrater agreement (Campion et al., 1993; James, 1982; James et al., 1984) was assessed using the multiple item estimator for within-group interrater reliability as proposed by James et al. (1984). With an average score across all teams of 0.87, this test yielded results indicating generally very strong agreement of ratings referring to the same team. Given this homogeneity of within-team ratings, data were aggregated by calculating the arithmetic mean. Furthermore, reliability analysis documents satisfactory internal consistency of the seven-item measure with a Cronbach's alpha of 0.86. Finally, the items were combined to measurement scales by calculating the arithmetic mean across the items pertaining to one construct.

For product quality, we asked foremen in prototype assembly and testing about the technical properties of the module a team was chartered to design and develop (e.g. dimensional integrity, durability, functionality, etc). We have calculated an unweighted index of the nine properties rated by these product quality experts. To measure the teams' budget performance we used project controlling data documenting the percentage to which a team exceeded (or undercut) its planned product target cost as well as its development budget. The teams' schedule performance is expressed as the project managers' evaluation of how many weeks a team is behind or ahead of the project schedule. Table I provides descriptive statistics and inter-correlations of all variables.

Control Variables

We included team size as a control variable in our analysis. Team size was reported to us by the team leaders. The size of a team is an important structural variable with potential influences on the quality of a team's collaborative task process and project success (Gladstein, 1984; Campion et al., 1993). Large team sizes make it more difficult for team members to interact with all other team members given the dramatic increase of (possible) individual links between team members as team size grows.

Given that the teams were divided up into eight sub-projects, we control for possible dependencies between observations from the same sub-project in our analysis. Specifically, we have regressed the independent variables and the dependent variables on sub-project and saved the standardized residuals, which were used as the basis for further analysis. This procedure effectively controls for all constant and unmeasured differences across the sub-projects that may explain differences in the variables and relationships investigated. Using a one-way ANOVA test, four of this study's main variables (i.e. project structuring and support at time 2, product quality, development cost, and development time) show significant mean differences across the eight sub-projects.

Data Analysis

The longitudinal multi-informant design of this study allows for temporal spacing of cause and effect. As described above, we are using measures of the independent variables (i.e. project structuring and support, team interface management) at time 1 and time 2, while we are considering team performance measures at time 3. We have conducted regression analyses in testing our causal hypotheses and t-tests for testing our hypotheses regarding differences in means across project phases. All analyses were conducted at the team level with pairwise exclusion in case of missing data. Collinearity statistics calculated for all regression analyses do

Table I. Descriptive statistics and intercorrelations

	<i>Time</i>	<i>Mean</i>	<i>St. Dev.</i>	<i>1</i>	<i>2</i>	<i>3</i>	<i>4</i>	<i>5</i>	<i>6</i>	<i>7</i>	<i>8</i>
(1) Project Structuring and Support	1	2.51	0.92								
(2) Team Interface Management	1	3.67	0.28	0.22 ⁺							
(3) Project Structuring and Support	2	2.33	0.62	0.42**	-0.31*						
(4) Team Interface Management	2	3.67	0.41	-0.25 ⁺	-0.11	0.03*					
(5) Product Quality (Expert Rating)	3	3.35	0.33	-0.23 ⁺	0.08	0.17	0.31*				
(6) Product Cost (% above target)	3	0.14	0.15	-0.33 ⁺	0.16	-0.25	0.36*	0.05			
(7) Development Cost (% above target)	3	0.10	0.25	-0.03	-0.23	-0.04	-0.01	-0.44**	-0.32*		
(8) Develop. Time (weeks behind schedule)	3	6.11	5.01	0.08	-0.06	-0.25 ⁺	-0.20	-0.03	-0.04	-0.31*	
(9) Team Size (number of team members)	-	9.33	3.30	0.33*	0.10	0.38*	-0.07	-0.06	0.00	-0.18	0.07

** = correlation significant at the 0.01 level.
* = correlation significant at the 0.05 level.
+ = correlation significant at the 0.10 level.

Table II. Regression analyses

		<i>Product Quality</i>	<i>Product Cost (% above target)</i>	<i>Development Cost (% above target)</i>	<i>Development Time (weeks behind schedule)</i>
<i>At the Project's Conclusion (Time 3)</i>					
Team Size		-0.29*	-0.12	0.11	-0.08
Project Structuring and Support	<i>Concept Phase</i>	-0.48**	-0.14	0.07	0.61**
Team Interface Management	<i>(Time 1)</i>	0.20	0.30	0.00	-0.62**
Project Structuring and Support	<i>Development Phase</i>	0.43**	0.23	0.04	-0.52**
Team Interface Management	<i>(Time 2)</i>	0.16	0.26	0.10	0.02
R-square		0.39	0.15	0.03	0.46
F		3.91	0.68	0.18	4.96
p		0.01	0.65	0.97	0.00

** = significant at the 0.01 level.

* = significant at the 0.05 level.

+ = significant at the 0.10 level.

not indicate distortions of results due to correlation among independent variables (variance inflation factor is below 1.5).

RESULTS

The regression results documented in Table II provide varying support to our causal hypotheses. Project structuring and support during the development phase increases product quality (std. coef. = 0.43) and reduces schedule slippage (std. coef. = -0.52). Surprisingly, this influence is reversed for the concept phase, where project structuring and support reduces quality (std. coef. = -0.48) and increases schedule slippage (std. coef. = 0.61). Product costs and development costs are not significantly influenced by project structuring and support in either phase. As such, these results offer only partial support to Hypothesis 1a based on the results for the development phase, but critically challenge the prediction regarding the effects of project structuring and support in the concept phase.

The results of the regression analyses partially confirm our Hypothesis 2a regarding the performance effects of team interface management in the concept phase. Team interface management in the concept phase decreases schedule slip-

page (std. coef. = -0.62) as measured at the project's conclusion. No significant results were obtained for product quality, product cost, and development cost. Furthermore, team interface management in the development phase shows no significant relationship with any of the four team performance measures, thus lending support to our prediction of team interface management being particularly important in the concept phase.

Relating to our Hypotheses 1b and 2b, we tested for possible differences in means between management functions performed in the concept and the development phases. T-tests (2-tailed) found no significant differences between the two phases. This indicates that, across all teams, the level of activity regarding the management of task interdependencies remained largely unchanged between the concept phase and the development phase. These results provide support for our Hypothesis 1b (predicting equally strong project structuring and support across both phases) but fail to support Hypothesis 2b (predicting more team interface management in the concept than in the development phase).

On a more exploratory note, the correlations between both phases are striking. While the perceptions of team interface management at both times are not significantly correlated, the evaluations regarding the performance of project structuring and support are significantly correlated. These findings illustrate a certain stability of the team leaders' perceptions of project structuring and support, thus suggesting that project managers tended to be less phase-specific in the performance of their managerial functions. The performance of team interface management at the early and the middle phases, however, seem to be largely unrelated, which in turn, may indicate that teams were exercising this managerial function more phase-specifically.

DISCUSSION

The main goal in this study was to investigate the management of task interdependencies in multi-team projects. The hypotheses that we derived received mixed support, confirming important propositions while challenging and questioning others. The following is a discussion of this study's main findings, highlighting how our results integrate and contribute to the literature. Furthermore, practical implications, the need for future research, and this study's limitations are also addressed.

It should be noted that our multi-informant longitudinal research design aimed at avoiding any common source bias while ensuring content validity. Consequently, the analyses conducted assess relationships between variables that are temporally spaced by 12 or 24 months, and are based on information from different respondents or sources. As such, this research design, compared to commonly used cross-sectional and single-informant research designs, imposes very stringent requirements, lending credibility to the empirical relationships found. Given this,

the regression models estimated document the importance of project structuring and support as well as team interface management for product quality and development time, with variances explained (R^2) of 39% and 46% respectively. Product cost and development costs seem less influenced by team-level or project-level managerial activities (R^2 of 15% and 3% respectively), with neither project structuring and support nor team interface management showing significant influences in either project phase. This, however, may have to do with budget considerations having been generally perceived by the project managers, and hence by the teams, as the less important of the four performance dimensions. In personal interviews project managers described the development budget as very important, but indicated that they would compromise budget performance for quality and time if necessary.

Theoretical Implications

Our study's results emphasize the importance of team interface management in the early project phase. This supports the findings of Ancona and Caldwell (1990) on team-external functions in single-team projects and shows that such boundary activities are also critical for schedule performance of teams in multi-team projects. Moreover, our findings are consistent with earlier arguments for single-team projects (Ancona and Caldwell, 1990; Gersick, 1988; Katz, 1982) and give evidence that it is the early concept phase of an innovative multi-team project where team interface management is most critical. Prior contributions have either failed to recognize time as a critical dimension in the successful application of managerial activity (Mitchell and James, 2001) or have stressed the early concept phase as particularly critical without providing the necessary empirical evidence (Gales et al., 1992; Kazanjian et al., 2000; Thamhain and Wilemon, 1975).

As this research documents, it is the concept phase where teams must identify related teams and develop good working relationships quickly to aid their module concept planning and their ability to ultimately achieve product quality targets while adhering to schedule objectives. When asked about the characteristics of successful module teams, the project director stated: 'The high performing teams clarify their relevant interfaces right in the beginning in a proactive way. After that, however, they focus on their design work. Teams that keep communicating across their boundaries will be overtaken by the other teams imposing their requirements on them.' As such, interface management should take place early in the project and is critical for negotiating technical arrangements between teams based on given technical interdependencies. After the first phase of figuring out all interdependencies, important relationships, and contact persons in other teams, many teams had found their way to deal with their interfaces. Those teams indicated that they 'systemized' their relationships with other teams and established routines. They were better able to judge what kind of information was important to others,

and how best to communicate (e.g. via email, hard copies of files, and/or personal conversations) to quickly resolve issues that have come up.

Contrary to our prediction, project structuring and support in the concept phase showed generally negative effects on team performance, while showing the expected positive influences as performed in the development phase. Though this is surprising given our prior hypotheses, we see a plausible explanation for the detrimental effect of project structuring and support in the concept phase on team performance (i.e. quality and time) at the project's conclusion. It may be that in the early phase, the teams better establish and arrange their interfaces among themselves, and project structuring and support may be counterproductive to this endeavor. Gerwin and Moffat (1997) as well as Kazanjian et al. (2000) note that interference in the teams' sphere by project managers can be destructive. As such, it is possible that project-level managerial activity aimed at identifying interdependencies and establishing corresponding inter-team relationships may be 'imposed' on the team. This may well be related to weaker team performance in the end, as such a top-down approach likely minimizes the team's role in shaping good working relationships with interdependent teams and possibly perturbs the team's self-confidence (Cusumano and Selby, 1995; Gerwin and Moffat, 1997). Moreover, it seems possible that strong project structuring and support in the concept phase also leads to a more manager-mediated form of coordination between interdependent teams, rather than directly initiated and driven by the teams themselves. Such more indirect coordination between teams, however, is likely inadequate given the highly uncertain and dynamic nature of the concept phase (Daft and Lengel, 1986; Hoegl et al., 2003).

The results from this investigation nevertheless confirm the proposed positive influence of project structuring and support in the development phase. This integrates with previous research indicating that project managers need to foster communication between teams, facilitate the resolution of conflicts, as well as again clarify, and where necessary adapt, the organizational structure and resource allocations to changing technical realities as the project unfolds (Kazanjian et al., 2000). It is in the development phase where the teams must focus on the implementation of their initial designs, where their attention turns to technical details and economic implications (e.g. product costs). As changes inevitably occur due to unforeseen problems (e.g. a preferred technical solution proves infeasible, test results are negative), the teams may need to adjust their designs under increasing cost and time pressures and revise prior commitments to other teams. In this phase, project management's role in providing structure and support becomes critical in helping the teams address and resolve possible conflicts at their interfaces and rearrange necessary resource allocations. From this perspective, the sharply differing effects of project structuring and support (i.e. negative in the concept phase, positive in the development phase) may reflect the different managerial requirements in the concept versus the development phase.

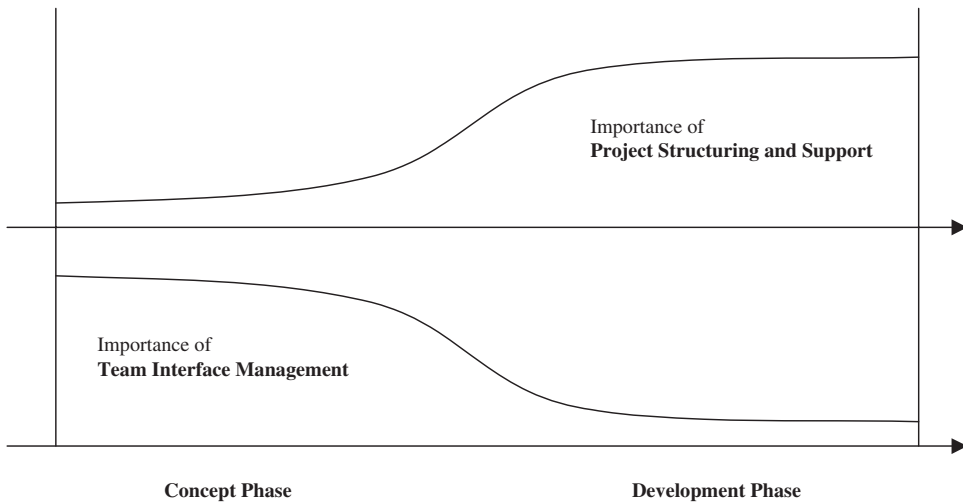


Figure 1. An illustration of the relative importance of team interface management and project structuring and support over the concept and the development phases

Taken together, the findings from this research highlight the importance of team interface management in the concept phase and project structuring and support in the development phase. It is the teams themselves that must directly identify interdependent teams and develop good working relationships in the concept phase, where direct coordination and cooperation are particularly important as the basic design of all modules and the overall product are being shaped. In the development phase, however, when the teams are busy with prototype development and testing as well as coordination with suppliers, project managers' structuring and support serves an important role in ensuring later team success. The teams' relationships with interdependent teams often increase in tension and conflict potential in the development phase due to their prior commitments to certain technical solutions. At the same time, the challenging nature of time and cost objectives become more apparent with increasing project duration as prototype testing may reveal expensive changes and initial schedule and cost slippage sets in. In this increasingly 'hectic and heated' environment, the teams are often unable to resolve conflicts at their interfaces and to find necessary trade-offs with other teams easily and quickly. As such, this study contributes to prior research in that it identifies when team-level and project-level managerial actions are most important for managing task interdependencies during the course of multi-team product development projects. Figure 1 illustrates the basic proposition stemming from this research by showing the relative importance of team interface management and project structuring and support over the course of the concept and development phases.

As shown by the non-significant differences between the project phases, the investigated management functions have been performed at the team and at the

project level on average to the same extent in the concept and the development phase. The non-significant correlation between team interface management performed in the concept and the development phases is not entirely surprising. It is plausible that, after performing this function in the concept phase, teams might lower their efforts in that area as they no longer recognize as much need for it in the development phase; or, on the contrary, teams neglecting boundary management in the beginning of the project may need to make up for that in the development phase. And yet other teams performed this function indeed to about the same extent in both phases. On the whole, these temporal relationships appear to outweigh each other. As indicated by the correlations between project structuring and support in both phases, there was not as much change in project structuring and support. Project managers remarked in our interviews that it was their intent to provide some kind of stability to the teams in this turbulent environment.

Practical Implications

What can managers learn from this study? Concerning the teams' task of managing their inter-team interfaces, our research shows that this function is important right from the beginning of a project. Therefore, team members who are not experienced in interface management should be prepared before the project starts, e.g. through training and education programs, to exercise this function. This will be more effective than a trial and error approach (Gemmill and Wilemon, 1994). Team members in this project complained that if they engaged too much in these tasks, they would have not enough time left for tackling actual engineering work. As the results from this study indicate, team members must be made aware of the importance of boundary management to team and project success, and educated before the project starts on how to exercise this function effectively and efficiently. These findings clearly point to the importance of management training on the team level, particularly in the technology and engineering dominated workforce of innovating organizations.

As for the project level, the results suggest that senior project managers promote team performance best by providing project structure and support while leaving enough room for autonomy and creativity to the teams, particularly in the early concept phase of an innovative project (Gerwin and Moffat, 1997; Imai et al., 1985). The project managers should act more as facilitators than as supervisors, and support the teams to find and structure their own relationships (Uhl-Bien and Graen, 1998). More guidance in terms of adapting and clarifying project structures and resource allocations is needed in the development phase of the project, when the module concepts are completed and the teams' need to focus inward in order to proceed with developing and testing prototypes.

Limitations and Future Research

Some limitations of this study are worth noting. First, the study included one multi-team project in the European automotive industry. While the hypotheses presented in this paper are not industry or country specific (rather, many of the arguments are derived from international scholarly literature), further investigations on this subject in other countries and industries are necessary to substantiate the empirical knowledge base in this field. Second, we investigated the effect of project-level and team-level management functions on team performance. Further studies should examine the link of management functions with *project* performance. This is only possible by including a large number of multi-team projects. Third, while we have (in the interest of providing external evaluations) surveyed team members on the team management function and team leaders on the project management function, it may also be interesting to compare and contrast our findings with self-assessments of both project and team leaders regarding their own project structuring and interface management activities. Finally, the project sequences as illustrated above portrait a somewhat rough and ideal picture of a product development project. As projects might encounter barriers or set-backs (Cusumano, 1997; Gales et al., 1992), a more detailed analysis of the course of the project could reveal iterative processes and corresponding managerial requirements. A qualitative case study research approach may help to further explain the dynamics of team-level and project-level managerial processes during the various project phases. The theoretical considerations and empirical evidence from this research may provide a starting point for such necessary further research.

APPENDIX

Questionnaire items

Project structuring and support	The project managers clarify boundaries and interdependencies within the project
	The project managers clarify boundaries and interdependencies with the main (line) organization
	The project managers define responsibilities and decision authorities of team leaders
	The project managers ensure that our team has all necessary resources (e.g. budget, human resources, etc.)
	The project managers support communication between the teams

Appendix *Continued*

Team interface management	Our team clarifies its boundaries and interdependencies with other teams within the project
	Our team defines its process outputs precisely
	Our team coordinates its process outputs with other teams
	Our team defines its process inputs precisely
	Our team coordinates its process inputs with other teams
	Our team precisely defines the module it is designing
	Our team clarifies the technical interfaces of its module with other teams
Product quality	The team's parts demonstrate good functionality (e.g. turning, opening, suspending)
	The team's parts demonstrate optical attractiveness
	The team's parts demonstrate tactile attractiveness
	The team's parts can easily be mounted onto the car (e.g. accessibility, fit)
	The team's parts demonstrate strong durability
	The team's parts are very close to being finished
	The team's parts performed well in driving tests
	The team's parts performed well in laboratory tests
	The team's parts performed well in crash tests

Note: These items are translations.

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